

Q1. A CMOS inverter has the following device parameters:

NMOS: $V_{Tn}=0.5V$, $\mu_n C_{ox}'=115\mu A/V^2$, $(W/L)_n=2$, PMOS: $V_{Tp}=-0.6V$, $\mu_p C_{ox}'=30\mu A/V^2$, $(W/L)_p=4$.

The supply voltage is $V_{DD}=2.5V$, and the output load capacitance is $0.5pF$. Ignore channel-length modulation and the intrinsic capacitance of the inverter (i.e., only consider the large load capacitance).

- Calculate the low-to-high and high-to-low delay times of the output signals.
- Determine the maximum operating frequency of the output signal.
- Calculate the dynamic power consumption at this maximum frequency.
- If the circuit must be re-designed so propagation delay times are reduced by 25%, determine the required transistor (W/L) for NMOS and PMOS. How does this affect dynamic power and power-delay product?

[12M]

Q2. A microprocessor consumes $0.4 mW/MHz$ when fabricated using a $180 nm$ process. With typical standard cells (gates), the area of the processor is $0.7 mm^2$. Assuming a $100 MHz$ clock frequency, and $1.8 V$ power supply, determine:

- What will be the area, power consumption, and the power density of the same processor if scaled to $120 nm$ technology, assuming the same clock frequency? (Use fixed voltage scaling and constant frequency)
- If the supply voltage in the scaled $120 nm$ part is reduced to $1.5 V$, what will be the power consumption and power density?
- How fast could the scaled processor in part (b) be clocked? What would be the power and power density at this new clock frequency?
- Power density is important for cooling the chip and packaging. What would be the supply voltage here to maintain the same power density as the original processor?

[12M]

Q3. Consider a resistive inverter with the following parameters: driver transistor W/L is $1.5/0.5$, load resistance $R_L=75k\Omega$ and an output load capacitor C_L of $3 pF$. Assume $V_{DD}=2.5V$, $K_n'=280\mu A/V^2$ and $V_{tn}=0.5V$. Ignore channel-length modulation and body effect and assume a long-channel device.

- Calculate t_{pLH} , t_{pHL} using the average current method
- Are the rising and falling delays equal? Why or why not?

[12M]

Q4. Consider four designs of a 6-input AND gate shown in Fig. 1. Using the Logical Effort method, where G is the path logical effort, P is the path parasitic effort, H is the path electrical effort, and N is the number of stages:

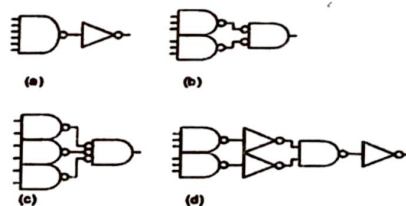


Fig. 1

- Calculate the delay D of each design (Fig1.a to Fig 1.d) when the path electrical effort is $H=1$.
- Suppose H is large enough ($H=20$), then identify which design provides the least delay.

[12M]

P.T.O.

Q5. For the depletion load NMOS circuit shown in Fig. 2, **determine $(W/L)_{Load}$** , such that the low level output voltage, V_{OL} does not exceed 0.4 V. Consider the following for the equivalent inverter circuit: Threshold voltage of the depletion load transistor (NMOS): $V_{T,Load} = -3$ V, Threshold voltage of the driver transistor (NMOS): $V_{T,Driver} = 1$ V. **[12M]**

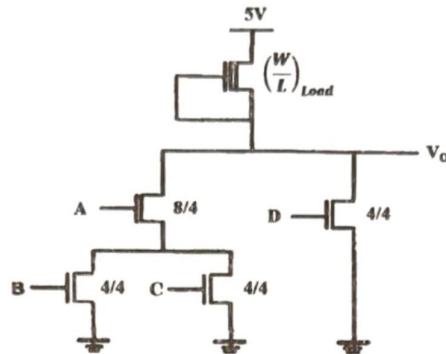


Fig. 2

Q6. Consider the layout shown in Fig. 3. Assume in this advance node active and implant are merged as a single layer.

- Draw the transistor-level schematic corresponding to the given layout, ensuring that all device connections are accurately represented.
- Derive and write the logic expression implemented by this circuit.
- Now, construct a differential logic gate using this expression (derived in part b).

[15M]

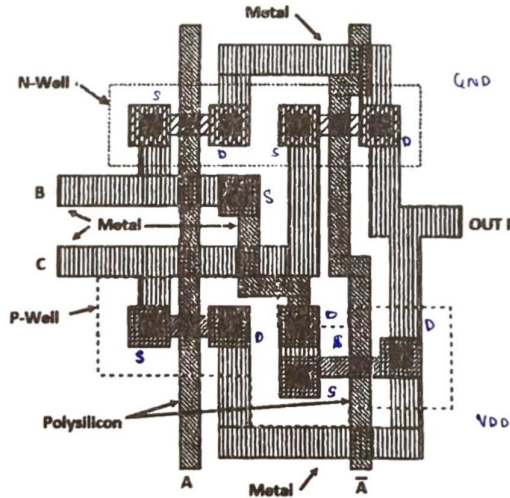


Fig. 3

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